FRAMES at four qubits: bounded local tuning

Separate n=4 calibration after n=2-tuned parameters failed the n=4 transfer gate

No joint-gate recovery winner: confirmation explicitly skipped.

Full physical design: d=16, 256 probes, 32 outcomes, 8,192 noiseless Born probabilities; rank one, Gell-Mann initialization, factor radius tau=4. Gate: RMSE <=1e-6, TPnorm <=1e-3, F >=0.99.

Candidate	alpha0	p	gamma(t), capped at 1	rho(t), capped at 1	Batch	Long nominee
incumbent	5e-05	0.25	4/(t+4)	4/(t+8)^(2/3)	32	—
penalty-div8	6.25e-06	0.25	4/(t+4)	4/(t+8)^(2/3)	32	—
penalty-div64	7.8125e-07	0.25	4/(t+4)	4/(t+8)^(2/3)	32	—
penalty-div512	9.765625e-08	0.25	4/(t+4)	4/(t+8)^(2/3)	32	—
penalty-slow	5e-05	0.125	4/(t+4)	4/(t+8)^(2/3)	32	—
step-wide	5e-05	0.25	16/(t+16)	4/(t+8)^(2/3)	32	—
step-slow	5e-05	0.25	(4/(t+4))^0.75	4/(t+8)^(2/3)	32	YES
batch128	5e-05	0.25	4/(t+4)	4/(t+8)^(2/3)	128	—
rho-half	5e-05	0.25	4/(t+4)	(8/(t+8))^0.5	32	—
penalty64-stepwide	7.8125e-07	0.25	16/(t+16)	4/(t+8)^(2/3)	32	—
penalty64-stepslow	7.8125e-07	0.25	(4/(t+4))^0.75	4/(t+8)^(2/3)	32	YES
penalty64-batch128	7.8125e-07	0.25	4/(t+4)	4/(t+8)^(2/3)	128	—

alpha(t)=alpha0(t+1)^p grows in every arm; beta(t)=1/[2alpha(t)]. The measurement gradient is in mean-loss units; summed-equivalent TP weight is lambda=8192 alpha, not batch × alpha. t counts optimizer updates, starting at zero.

Screen: 12 settings × channels 0/1, 1.536 million sampled rows each. Long: two nominated settings × the same channels, 12.8 million rows each, compared with the archived same-seed incumbent. This partial grid does not establish globally optimal tuning.

All screen outcomes, paired by channel

Each cell lists channel 0 then channel 1 | Same initializer/stream within channel | All 12 configurations retained

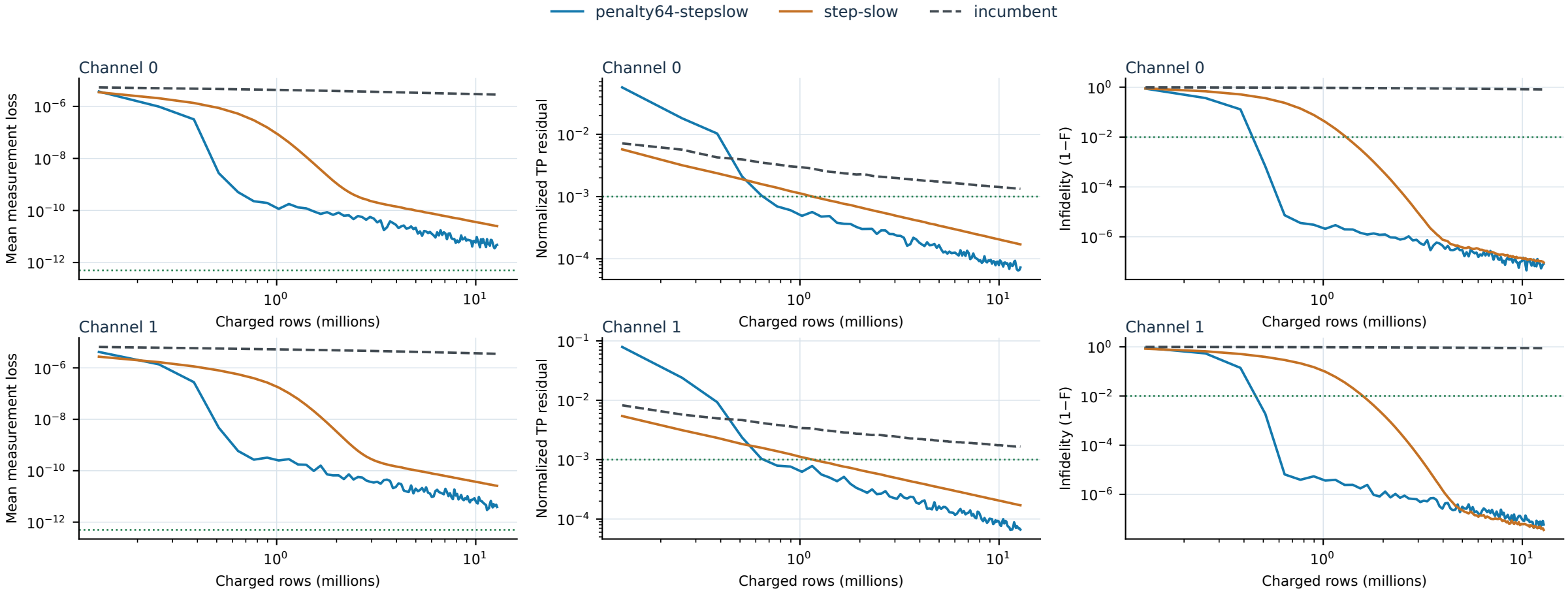
Configuration (* nominated)	RMSE c0 / c1	TP norm. c0 / c1	Infidelity (1−F) c0 / c1	Relative RMSE c0 / c1	Joint gate c0 / c1
incumbent	2.85e-03 3.16e-03	2.44e-03 2.97e-03	9.30e-01 9.59e-01	6.74e-02 7.45e-02	FAIL FAIL
penalty-div8	3.06e-03 2.73e-03	9.57e-03 9.30e-03	9.97e-01 9.33e-01	7.23e-02 6.45e-02	FAIL FAIL
penalty-div64	3.27e-03 2.87e-03	4.66e-02 3.62e-02	9.84e-01 9.27e-01	7.72e-02 6.76e-02	FAIL FAIL
penalty-div512	1.98e-03 2.42e-03	8.01e-02 1.33e-01	6.34e-01 8.68e-01	4.67e-02 5.71e-02	FAIL FAIL
penalty-slow	2.82e-03 3.17e-03	6.15e-03 7.69e-03	9.08e-01 9.79e-01	6.65e-02 7.48e-02	FAIL FAIL
step-wide	2.54e-04 4.49e-04	5.15e-04 6.67e-04	1.44e-02 5.25e-02	6.01e-03 1.06e-02	FAIL FAIL
step-slow *	1.13e-04 2.05e-04	8.14e-04 8.09e-04	3.27e-03 1.22e-02	2.67e-03 4.83e-03	FAIL FAIL
batch128	3.26e-03 2.67e-03	3.76e-03 2.91e-03	9.81e-01 9.19e-01	7.71e-02 6.29e-02	FAIL FAIL
rho-half	3.24e-03 3.49e-03	6.30e-03 7.15e-03	9.81e-01 9.95e-01	7.66e-02 8.22e-02	FAIL FAIL
penalty64-stepwide	1.08e-05 8.93e-04	3.57e-04 5.62e-03	6.85e-07 2.22e-01	2.55e-04 2.11e-02	FAIL FAIL
penalty64-stepslow *	1.36e-05 1.40e-05	3.78e-04 4.34e-04	1.44e-06 1.75e-06	3.21e-04 3.30e-04	FAIL FAIL
penalty64-batch128	2.96e-03 3.11e-03	4.07e-02 4.73e-02	9.83e-01 9.76e-01	6.98e-02 7.33e-02	FAIL FAIL

Relative RMSE = $||A(UU^H)-y||_2 / ||y||_2$. Nomination uses one smallest worst-channel joint deficit and one largest worst-channel fidelity; duplicates are filled by deficit order. Truth fidelity is used only for calibration selection.

Joint deficit D=max(RMSE/1e-6, TPnorm/1e-3, max(0,1−F)/0.01). Numerical failures remain visible and cannot be nominated. Penalized objectives with different alpha are never ranked.

Long-budget trajectories versus archived n=4 control

Two matched calibration channels | Same physical data, initialization and declared sampling seed | Fresh starts



Configuration	Chan.	RMSE	TP norm.	Relative RMSE	1−F	Gate	Persistent gate rows
penalty64-stepslow	0	3.09e-06	7.25e-05	7.31e-05	8.23e-08	FAIL	—
penalty64-stepslow	1	2.79e-06	6.63e-05	6.59e-05	5.96e-08	FAIL	—
step-slow	0	7.07e-06	1.71e-04	1.67e-04	9.34e-08	FAIL	—
step-slow	1	7.19e-06	1.71e-04	1.70e-04	3.59e-08	FAIL	—
incumbent	0	2.39e-03	1.33e-03	5.64e-02	8.13e-01	FAIL	—
incumbent	1	2.66e-03	1.65e-03	6.26e-02	8.80e-01	FAIL	—

Dotted: physical gate. Zero-row points omitted; plotted values floored at 1e-16. Crossings certify recorded checkpoints only.

Confirmation, total work and remaining limits

Eligibility requires both calibration channels to pass all three unchanged physical thresholds

CONFIRMATION SKIPPED: no configuration passed both calibration channels.

The two long nominees and the archived control were assessed by the same joint gate. No recovery winner is declared, and channels 2/3 were not used to rescue or retune the selection. Failed gates are completed scientific outcomes, not missing experiments.

This bounded grid can show progress or failure within its declared work budget; it cannot prove that FRAMES is incapable of solving this physical setup or that one penalty mechanism caused the failure.

Stage	Trials	Complete	Num. failures	Gate passes	New rows (M)	New updates
pilot	3	3	0	0	0.192	4,500
screen	24	24	0	0	36.864	1,008,000
long	4	4	0	0	51.200	1,600,000
confirmation	SKIPPED	—	—	—	0	0
TOTAL NEW	31	31	0	0	88.256	2,612,500

Archived control: 25.6 million rows from the previous round, zero additional compute here. All new starts, resource pilots and repeated prefixes are included above. Pilot gate outcomes are short diagnostics, not recovery tests.

Interpretation limits: larger batches mean fewer updates at matched row budgets and different terminal alpha; the sampling contrast is not an isolated variance test. Row counts exclude TP, LMO and metric work, so they are not runtime/FLOP equivalence. Seeds and native source are verified; realized sampling batches were not archived.

Only noiseless, rank-one channels are tested. Channels 1000–1019 remain reserved. The earlier n=2 result is not evidence of tuned n=4 recovery. Published Quiroga remains unresolved: this is FRAMES calibration, not a completed best-for-best comparison or a speed claim.

Validation: 133 passed in 3.65s. Source 4ce1108a0692. All input/output SHA-256 hashes are in the companion evidence JSON.